Module 2 Project Rubric

RiskLens | RAG Evidence Engine

C2 - Grounded Retrieval and Generation
Build and evaluate a retrieval-augmented evidence engine that finds relevant product information, supports its answers with auditable citations, refuses unsupported requests, and justifies retrieval and graph-based design choices using controlled measurements.

Evidence to prepare

- **D2.1:** At least 100 real reviews with provenance, the four CAPSTONE_DOCS unchanged with their source IDs, and at least two authored documents with planted facts and a cross-document link.

- **D2.2:** At least 20 frozen gold questions, including at least three multi-hop questions, with required sources and reference answers; authored before indexing.

- **D2.3:** A working retrieve(query, k=4) returning source, text, kind and score; recall@4 exceeds retrieve_v0 on the same gold set.

- **D2.4:** A working rag_answer() returning the answer and actual retrieved hits, supported citations, and refusal on at least four of five preselected unsupported questions using REFUSAL_PHRASES.

- **D2.5:** Faithfulness and answer-relevance results on the gold set, with judge identity, protocol, examples and bias caveats.

- **D2.6:** Saved Stage 2 evaluation: baseline/final recall table and refusal rate, with the corresponding metrics ledger entry.

- **D2.7:** One measured upgrade beyond the query prefix, with a before/after recall table and adopt/reject conclusion.

- **Design note:** One relational question, an explanation of how graph-assisted retrieval could help, and the costs and limitations of that choice.

How grading works

All 5 criteria must be **Met** to complete C2. **C2.3 is MUST PASS** and is reviewed first. If it is Not Yet, revise before the remaining criteria are reviewed. No points, weights or bonus credit.

Submit the same notebook link with saved outputs, a milestone snapshot, a short summary and known issues. Revise Not Yet criteria and affected evidence. The full RiskLens Grading Rubric defines review rules; course policy governs attempts and timing.

Module 2 | Assessment criteria (1/2)

C2 - Grounded Retrieval and Generation

RiskLens Criterion Set v2.0 | All 5 criteria must be Met. Standards are identical to the consolidated rubric.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C2.1 Your corpus and gold questions are valid	D2.1 CORPUS; D2.2 GOLD; source provenance and frozen snapshot.	Include at least 100 real reviews, all four official documents verbatim with their IDs, and at least two authored documents containing testable facts and a cross-document link. Before indexing, freeze at least 20 corpus-based gold questions, including three multi-hop cases, with required source IDs and reference answers. Preserve the version record.	Invented reviews presented as real; altered official facts; gold questions chosen after retrieval trials; multi-hop labels on single-source questions.
C2.2 Your retriever improves under a fair comparison	D2.3 retrieve(); D2.6 evaluation; D2.7 upgrade experiment.	Beat retrieve_v0 on recall@4 using the same corpus, frozen gold questions and metric implementation. Save baseline/final recall at k = 1, 2, 4 and 8. Measure one upgrade beyond the query prefix on unchanged questions, show before/after results, and justify adopting or rejecting it. A rejected upgrade is valid evidence.	Changing the gold set or corpus between arms; claiming recall@8 beats baseline recall@4; reporting only the prefix fix; omitting a negative experiment.
C2.3 Your answers are grounded and refuse unsupported claims MUST PASS	D2.4 rag_answer(); answer/hit pairs; five-question adversarial set and saved responses.	Return the answer and the actual retrieved hits. Citations resolve to those source IDs and support the associated claims. Preselect five genuinely unanswerable questions; at least four receive a refusal containing a REFUSAL_PHRASES phrase without an invented answer. Report all five outcomes; separately report any additional adversarial cases.	Decorative or nonexistent citations; refusing and then inventing an answer; changing adversarial questions after seeing failures; counting a phrase match as sufficient proof.

Evidence note. Assess saved outputs and linked artifacts; graders do not need to repeat model training. A missing evaluation output does not demonstrate a result. Preserve real prompts, configurations, data versions and outputs as the work happens. Dated snapshots or version history support freezing; cell position alone does not prove chronology.

Module 2 | Assessment criteria (2/2)

C2 - Grounded Retrieval and Generation

RiskLens Criterion Set v2.0 | All 5 criteria must be Met. Standards are identical to the consolidated rubric.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C2.4 Your generation evaluation measures both axes	D2.5 scoring outputs; D2.6 evaluation and ledger; judge protocol.	Score faithfulness and answer relevance separately over the frozen gold set. State the judge or evaluation tool, scoring instructions, sample size and aggregation, and preserve representative answer-level results. Explain judge bias, especially shared generator/judge weights. Save the Stage 2 outputs and ledger; reference-substring matches alone do not establish faithfulness.	Only measuring retrieval; using answer-substring matches as a faithfulness score; unnamed judge; no caveat about self-evaluation or judgment limits.
C2.5 Your retrieval design choice is justified	Design note beside D2.7; one question and its linked source IDs.	Trace a question requiring linked facts across documents. Explain whether graph-assisted retrieval would help and identify its extraction, indexing and maintenance costs. Distinguish the notebook graph-hop illustration from full GraphRAG and KAG systems. Justify the retrieval approach used in RiskLens; implementing a complete graph framework is not required.	Calling any vector index GraphRAG; presenting a toy graph as a complete KAG system; claiming graphs improve every query; no concrete linked question.

Module handoff. The starter counts a gold question as a retrieval hit only when every required source is retrieved. Retain that all-required-sources definition for baseline/final comparisons and label it clearly. Citation support and refusal meaning still require inspection beyond phrase matching.

Source mapping: D2.1-D2.7. RiskLens Criterion Set v2.0 governs criteria; the published course policy governs administrative rules.